

Telco Customer Churn Analysis

Turning Data Into Retention Strategy

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Dataset Overview

7,043

Total Customers

21

Features

26.5%

Churn Rate

0

Duplicates

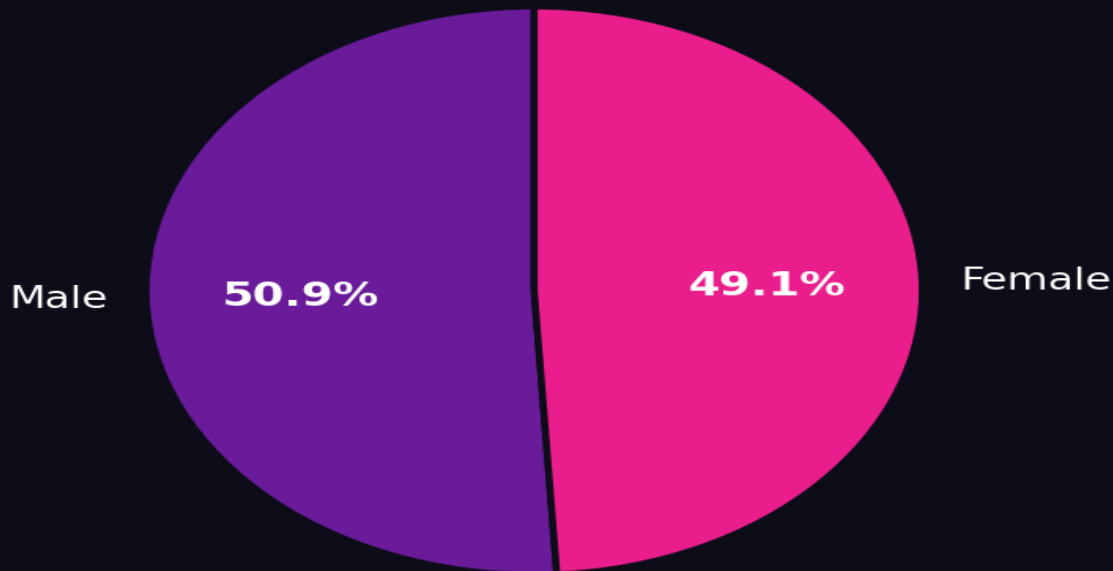
Data Cleaning & Preparation:

- TotalCharges stored as text — converted to numeric; 11 blank entries set to NaN
- Zero duplicate records found across all 7,043 rows
- Tenure segmented into four bands: 0–12, 13–24, 25–48, and 49–72 months
- Churn (Yes/No) encoded as binary 1/0 for correlation analysis

Customer Base Analysis

Customer Base by Gender

Customer Base by Gender



Key Insight

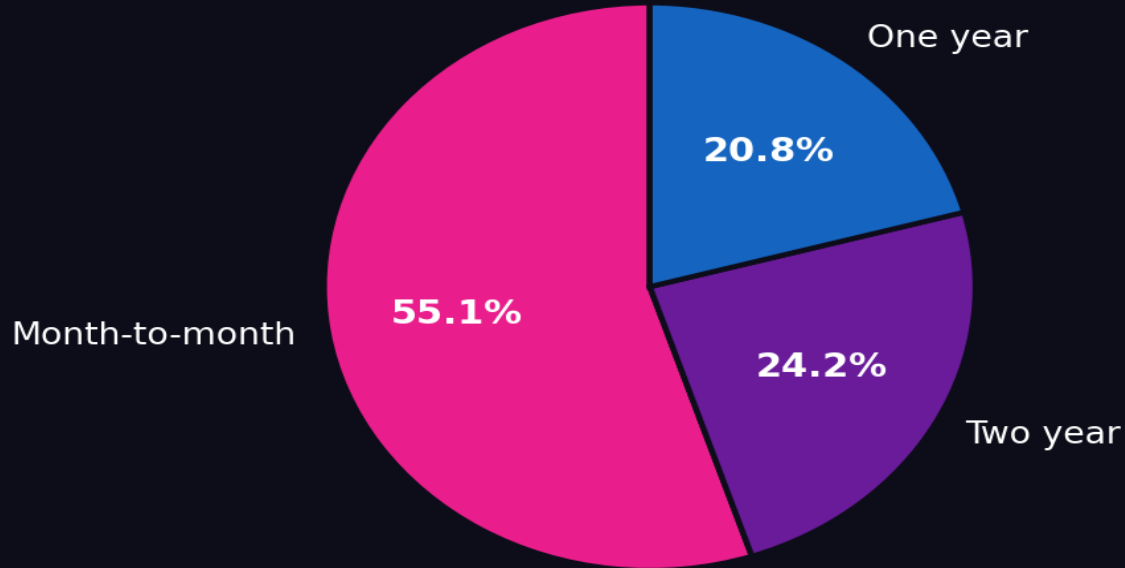
The base is almost evenly split — male 50.5%, female 49.5%.

Gender shows no meaningful difference in churn rate

Retention strategy should focus on behavioural factors, not gender demographics.

Customer Base by Contract Type

Customer Base by Contract Type



Key Insight

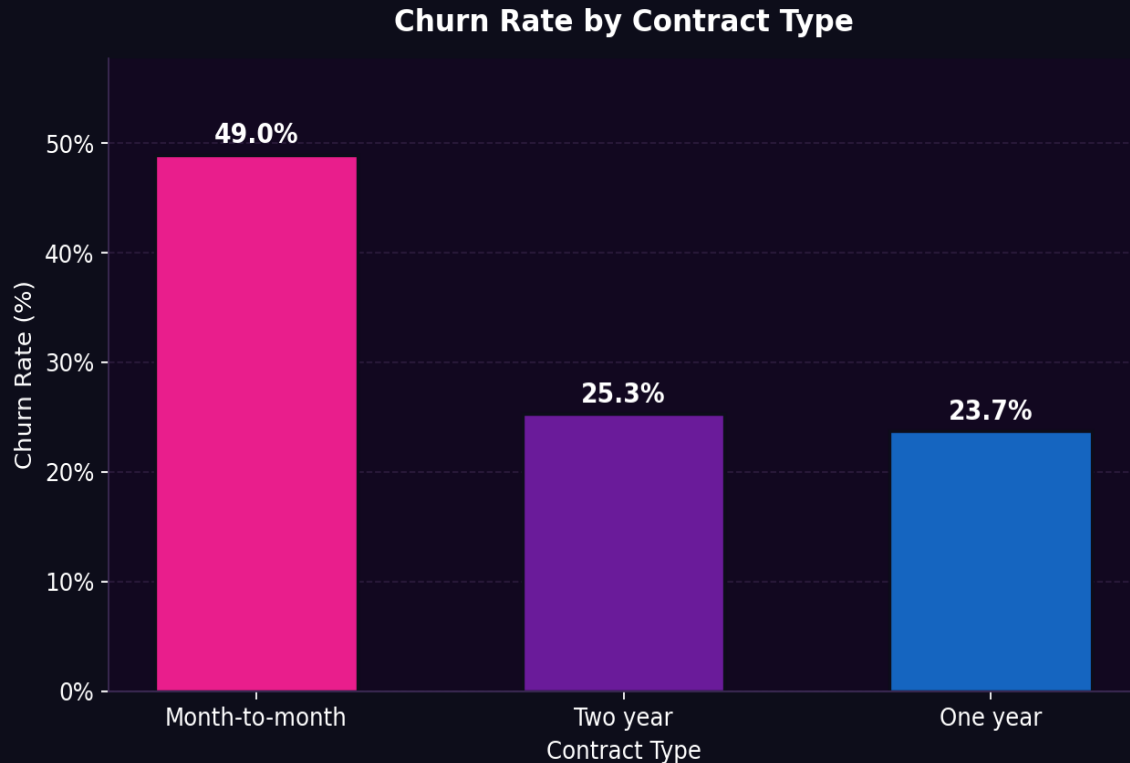
Month-to-month contracts make up ~55% of the customer base.

This is the highest-risk contract group meaning the majority of customers are already in the danger zone.

This is the single biggest retention lever available.

Churn Rate Analysis

Churn Rate by Contract Type



Key Insight

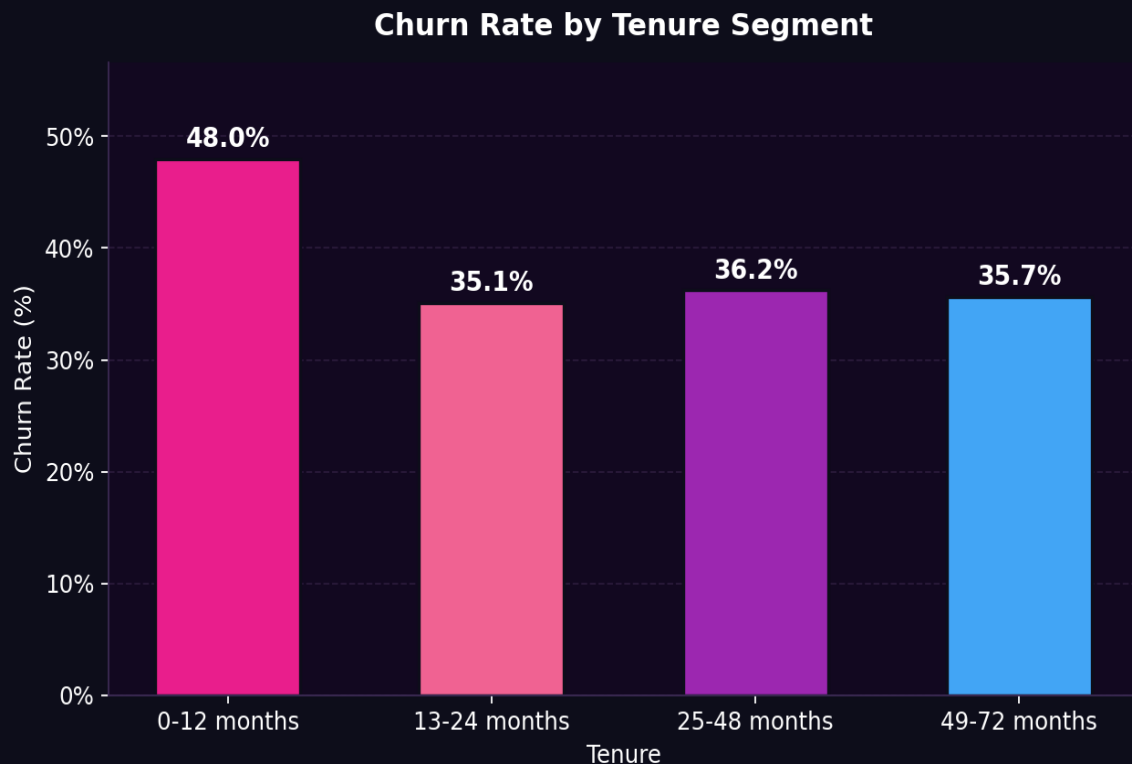
Month-to-month customers churn at ~49% more than twice the rate of one- and two-year contracts.

Longer contracts create switching barriers and signal commitment.

Business Risk: Most customers are on the highest-churn contract.

Opportunity: Incentivise upgrades to 1- or 2-year contracts.

Churn Rate by Tenure Segment



Key Insight

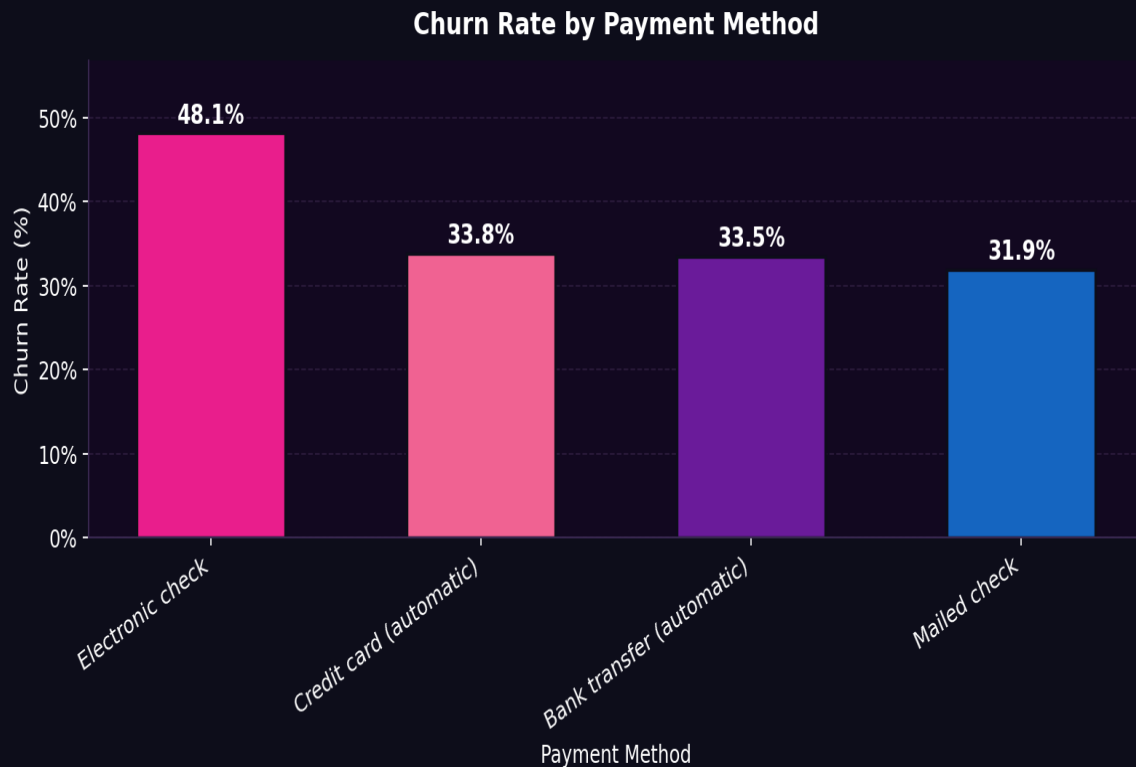
Customers in their first 12 months have the highest churn (~48%).

Churn falls steadily as tenure increases reaching ~11% at 49–72 months.

Once customers stay, they are far less likely to leave.

Business Risk: First year is the critical window nearly 1 in 2 new customers leave.

Churn Rate by Payment Method



Key Insight

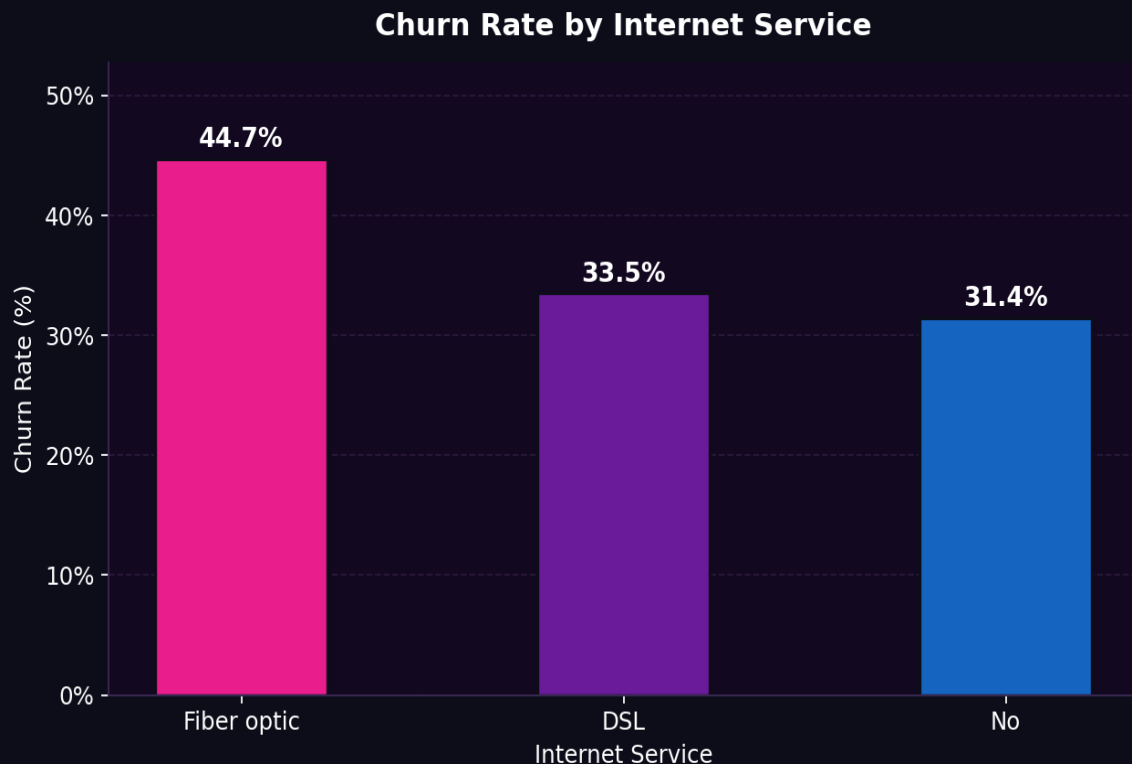
Electronic check customers churn at ~45% — nearly 3× the rate of automatic payment methods (~16%).

Manual payment methods both show elevated churn.

This may reflect billing friction, payment failures, or disengaged customers.

Opportunity: Incentivise migration to auto-pay methods.

Churn Rate by Internet Service



Key Insight

Fiber optic customers churn at ~42% more than twice the DSL rate (19%) and almost 6× no-internet (7%).

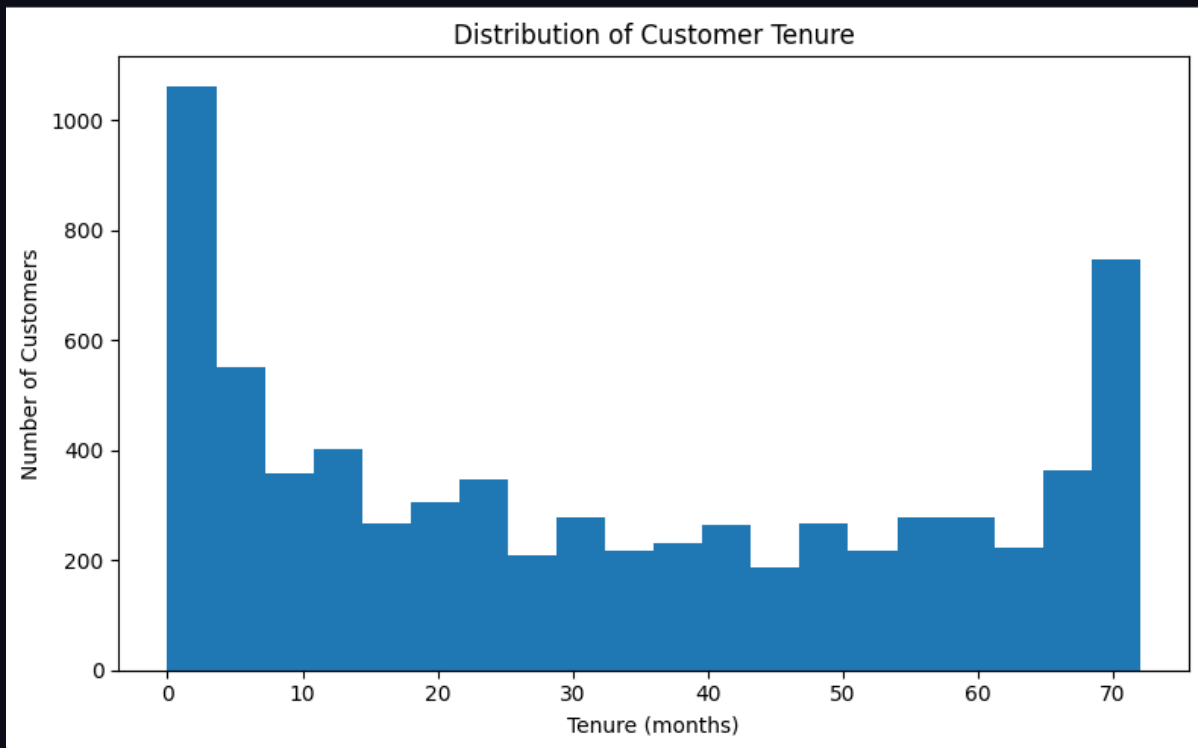
Despite being a premium, higher-cost service, fiber optic customers are far more likely to leave.

This signals a pricing, value, or service quality issue.

Opportunity: Investigate and fix the fiber optic customer experience.

Distributions & Monthly Charges

Distribution of Customer Tenure



Key Insight

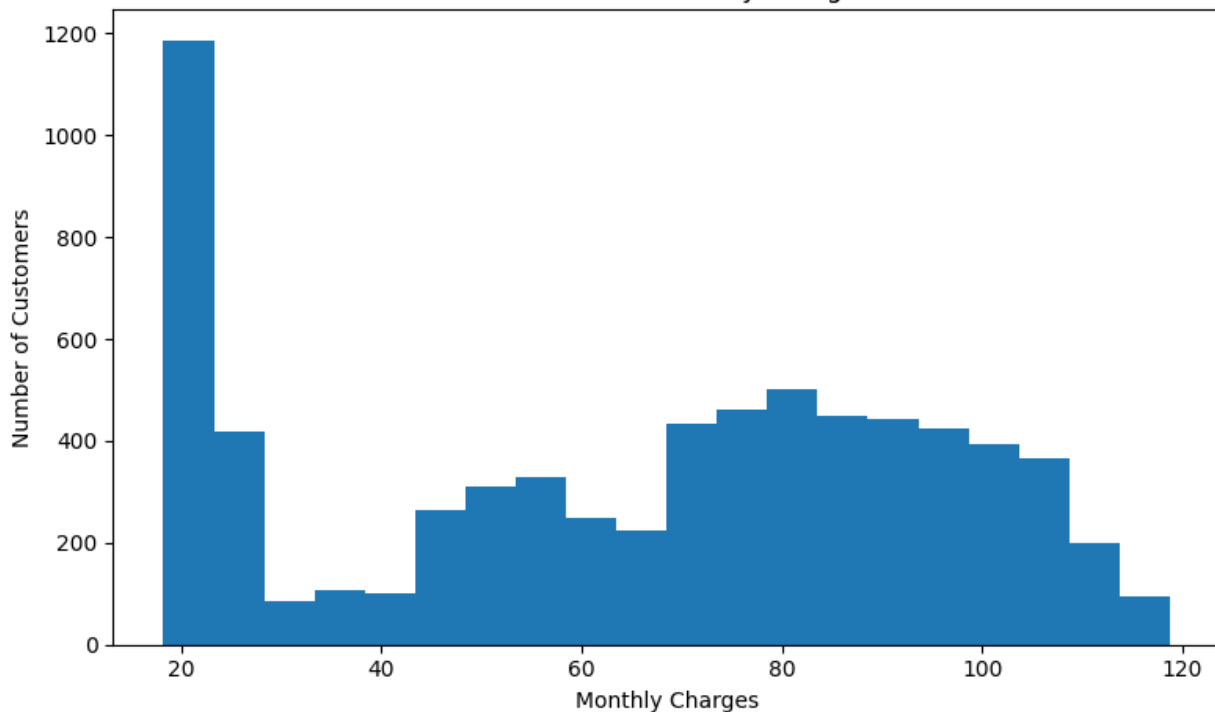
Two clear peaks: a large group of new customers (0–12 months) and a stable long-term group (60–72 months).

High new-customer volume, combined with high churn, amplifies total revenue impact.

The long-tenure group represents the loyal base — model their behaviours for retention learnings.

Distribution of Monthly Charges

Distribution of Monthly Charges



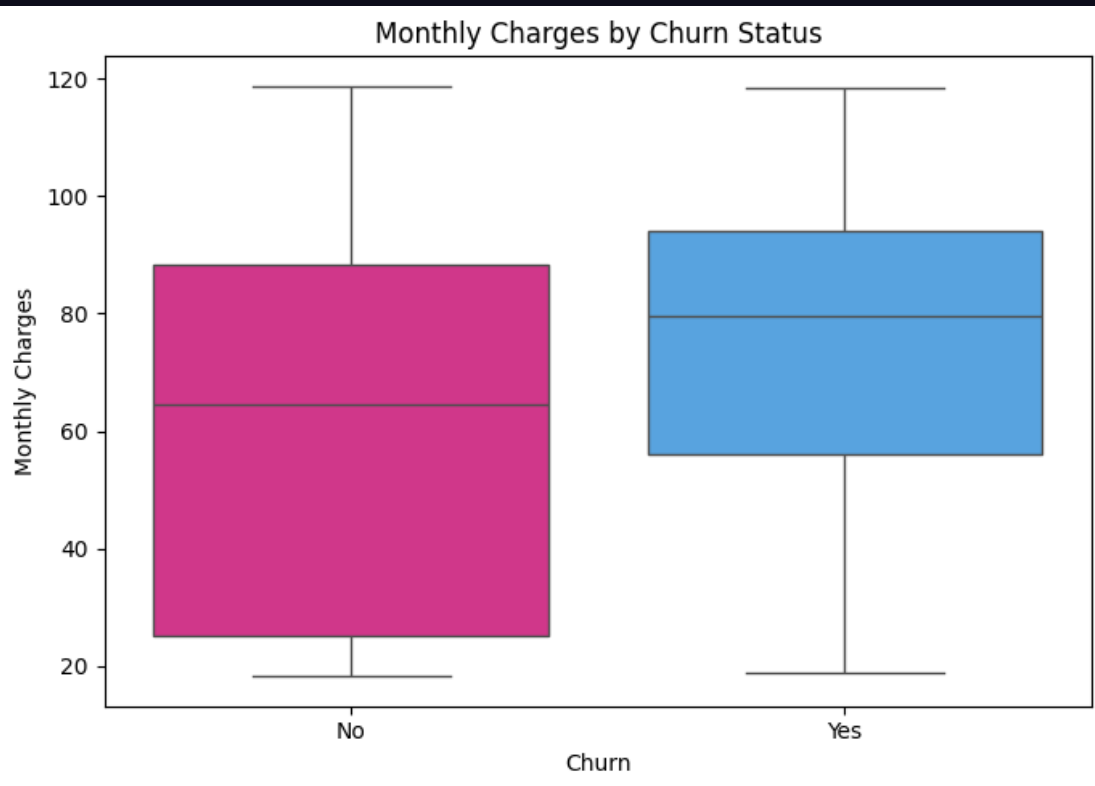
Key Insight

Monthly charges range broadly (\$18–\$119), with concentrations at lower-tier (~\$20–\$30) and higher-tier (~\$70–\$100) plans.

The bimodal pattern suggests two distinct customer segments by service level.

Higher-charge customers (fiber optic, multiple services) are among the highest-churn groups — a value perception gap.

Monthly Charges by Churn Status



Key Insight

Churned customers have a higher median monthly charge than those who stayed.

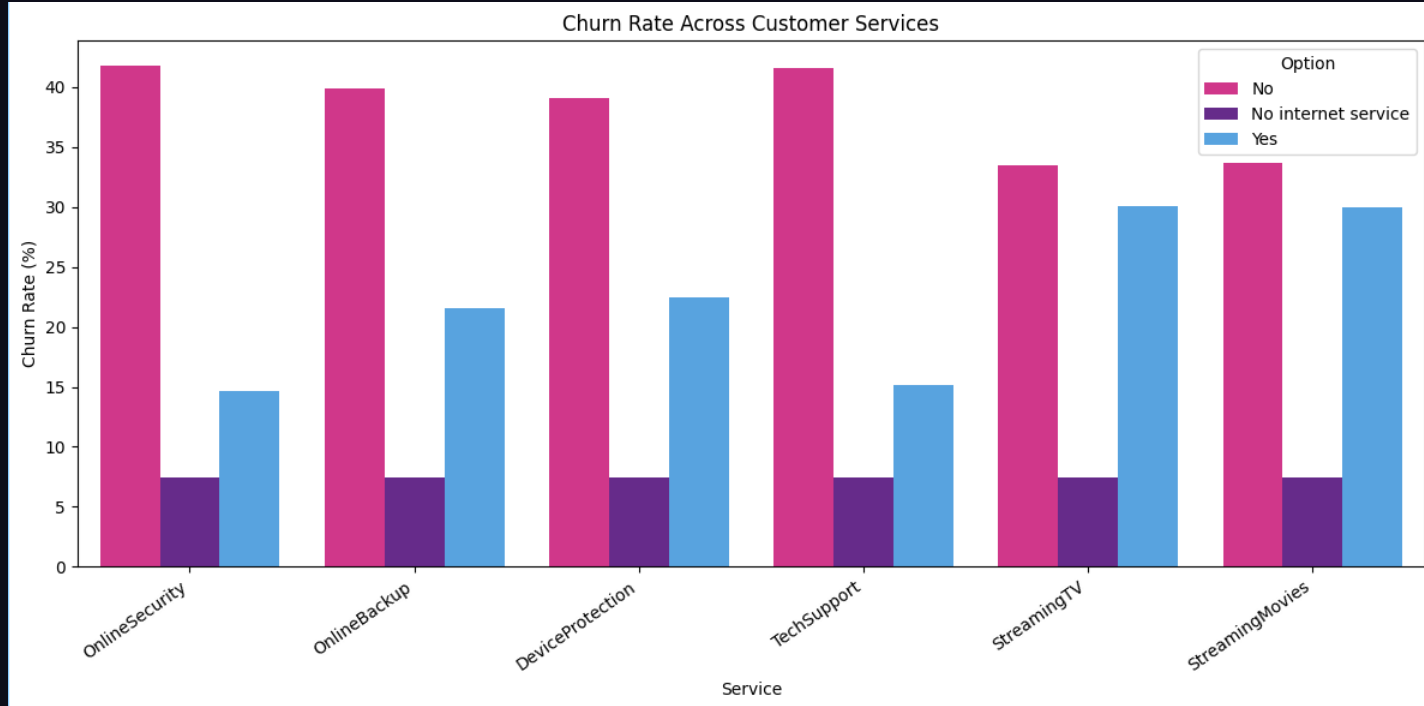
Customers on more expensive plans are disproportionately more likely to leave.

This reinforces the value gap: paying more does not mean perceived value matches cost.

Correlation: Monthly charges show a +0.19 correlation with churn.

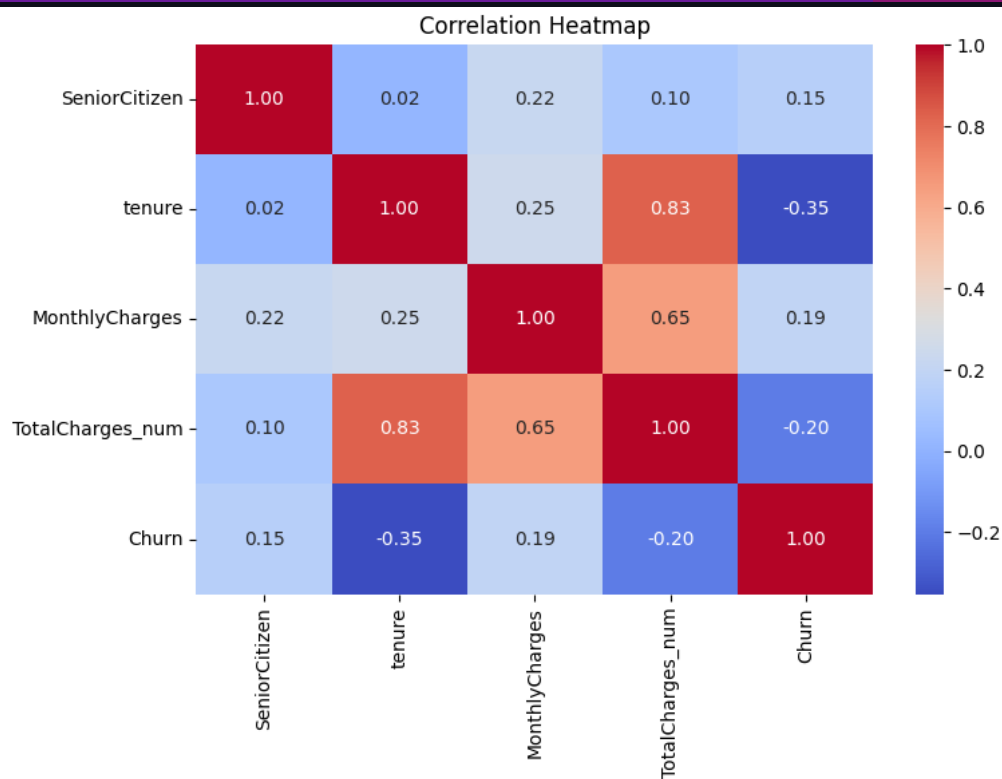
Services & Correlation Analysis

Churn Rate Across Customer Services



Key Insight: Customers without Online Security, Tech Support, and Online Backup consistently show higher churn. This is an association, not proven causation — but service bundles may improve perceived value and retention.

Correlation Heatmap



Tenure → Churn

-0.35

Strongest negative link — longer-tenure customers are far less likely to churn.

Monthly Charges → Churn

+0.19

Positive link — higher-paying customers churn more (value gap).

Total Charges → Churn

-0.20

Reflects tenure effect: longer tenure = more total spend = lower churn.

Senior Citizen → Churn

+0.15

Seniors churn at above-average rates across all segments.

Business Insights, Risks & Recommendations

Five Business Insights

1 Month-to-month customers churn at ~49%, more than twice the rate of one- or two-year contracts.

2 New customers (0–12 months) churn at ~48%, the first year is the critical retention window.

3 Electronic check users churn at ~45%, far above automatic payment methods (~16%).

4 Fiber optic customers churn at ~42% versus ~19% for DSL, despite paying a premium.

5 Senior citizens, customers without partners, and those without dependents all show above-average churn.

Business Risks & Opportunities

3 Business Risks

Revenue Erosion

High month-to-month churn threatens recurring revenue — most customers are in this highest-risk group.

First-Year Gap

Nearly half of new customers churn within 12 months; onboarding is failing to build loyalty.

Segment Pain Points

Fiber optic and electronic-check groups churn at 2–3× the baseline — pain points are unresolved.

3 Business Opportunities

Contract Upgrade Incentives

Targeted offers to migrate month-to-month customers to 1- or 2-year contracts stabilise revenue.

First-Year Retention Programme

Structured onboarding and early offers in months 1–12 can significantly reduce early churn.

Fix High-Churn Segments

Improve the fiber optic value proposition and simplify electronic-check billing.

Five Actionable Recommendations

1

Target Month-to-Month Customers

Offer discounts, perks, or loyalty rewards to migrate customers to 1- or 2-year contracts.

4

Review the Fiber Optic Experience

Audit pricing, service quality, technical support, and customer complaints in this segment.

2

Focus on the First 12 Months

Structured onboarding, proactive outreach, and early-retention offers for new customers.

5

Promote Support & Security Services

Bundle Online Security, Backup, and Tech Support for at-risk customers.

3

Investigate Electronic Check Churn

Understand billing friction and payment failures — then incentivise switching to auto-pay.

Final Conclusion

Churn is not evenly distributed. The strongest patterns centre on contract type, tenure, payment method, and internet service.

Management should direct retention resources toward the highest-risk segments, month-to-month customers in their first year, electronic-check payers, and fiber optic subscribers, rather than treating every customer the same.

Targeted, data-driven retention, not broad-based spend, is the path to meaningful churn reduction and more stable recurring revenue.

**Contract
Strategy**

**First-Year
Retention**

**Payment
Experience**

**Service
Quality**